

Problem 1: Put a penny on each gray square of strip A. The thick line is the river. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump so that a peg crosses the river. Write an X in the square where it lands.

Strip A

Problem 2: Strips B, C and D start with three, four and six pegs on the home side of the river. Take turns: one of you jumps while another records, and swap after each strip. For each strip, put pennies on the gray squares, then jump as many times as you like, in either direction, and try to get a peg onto square 2, the square with the star. Write the farthest square any peg reached, 1 or 2, in the blank above the strip. On strip E, draw a dot in up to six home squares to make a starting position for the person next to you. Swap pages, play the position you were given, and write the farthest square reached.

Strip B

farthest square reached: _____

**Strip C**

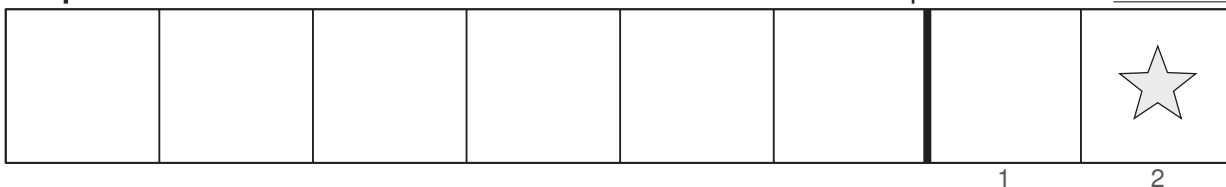
farthest square reached: _____

**Strip D**

farthest square reached: _____

**Strip E**

farthest square reached: _____



Problem 3: Strip F has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip F

1	1	2	3	5	8	13	21
							
						1	2

pegs on the squares valued . . .	total before	total after	up, same or down
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all six home squares; make the only jump that is possible	_____	_____	_____
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2 and 3; jump to the left	_____	_____	_____
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2 and 3; jump to the right	_____	_____	_____
----------------------------	-------	-------	-------

the second 1 and 2; jump to the left	_____	_____	_____
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8 and 13, one on each side of the river; jump to the left	_____	_____	_____
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Problem 4: Use strip F again. Try to find a jump anywhere on the strip that makes the total go up: start from any two pegs you like, and try jumps to the right and jumps to the left. Write three of your tries in the table, then circle your answer: did any jump make the total go up? yes no. When a peg jumps to the right it leaves two squares and lands on the next one. Explain to the adult at your table why the total is exactly the same after every jump to the right, and why every jump to the left makes it go down.

from the square valued . . . over . . . to . . .	total before	total after	up, same or down
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_____ over _____ to _____	_____	_____	_____
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_____ over _____ to _____	_____	_____	_____
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_____ over _____ to _____	_____	_____	_____
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Problem 5: Write the total of the six home squares of strip F: _____ and the value of square 2: _____. Explain to the adult why no peg can ever land on square 2 of strip F, whatever jumps are made and whichever home squares the pegs start on. Then write the reason in two or three sentences.

Problem 6: Strip G has seven home squares and no values. Write 1 and 1 on its two leftmost squares. Then choose the other values yourself so that a jump toward the star never changes the total. Check that every jump away from the star lowers it. Write the total of the seven home squares: ____ and the value of square 2: _____. On strip F the home squares were worth 20 and square 2 was worth 21. Work out the same two numbers for a strip with five home squares: ____ and _____, and for a strip with eight home squares: ____ and _____. Tell the adult why the army is always exactly one short, and why that will still be true for a strip of any length.

Strip G

									★
								1	2

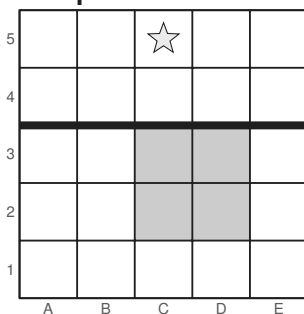
Problem 7: Board O is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2. Cross the river with one jump. Write the name of the square where the peg lands: _____

Board O

			★		
5					
4					
3					
2					
1					
	A	B	C	D	E

Problem 8: Each small picture shows a shape of four pegs on the home side of board O. Share the six shapes out around the table, two each. For each of your shapes, copy it onto board O with pennies and try to get a peg to the star at C5, jumping as many times as you like. Under the picture write *yes* if a peg reached the star and *no* if it never did. For a shape that works, also write its jumps in order, like C1 over C2 to C3. Before the table writes *yes* in the table results, a different person must replay those jumps on board O.

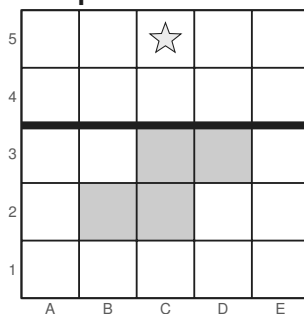
Shape I



works: _____

jumps: _____

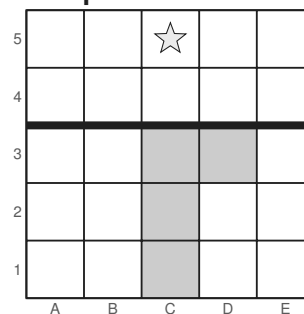
Shape J



works: _____

jumps: _____

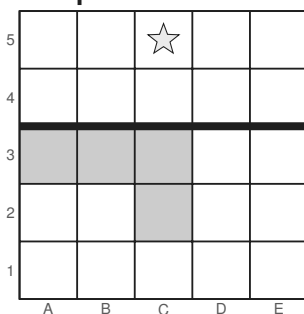
Shape K



works: _____

jumps: _____

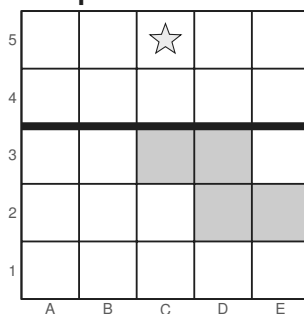
Shape L



works: _____

jumps: _____

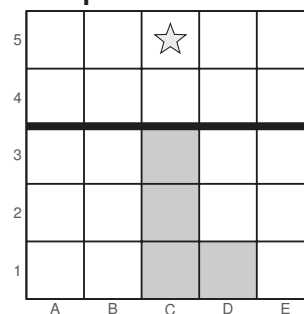
Shape M



works: _____

jumps: _____

Shape N



works: _____

jumps: _____

table results	Shape I	Shape J	Shape K	Shape L	Shape M	Shape N
reached the star (yes or no)						

Problem 9: Board P shows a value in the corner of every square of board O. For each line, put pennies on the squares named, make the jump, and write the total before and the total after.

C3 and C2; jump up	before _____	after _____
E3 and D3; jump to the left	before _____	after _____
B3 and C3; jump to the right	before _____	after _____
C4 and C3; jump down	before _____	after _____

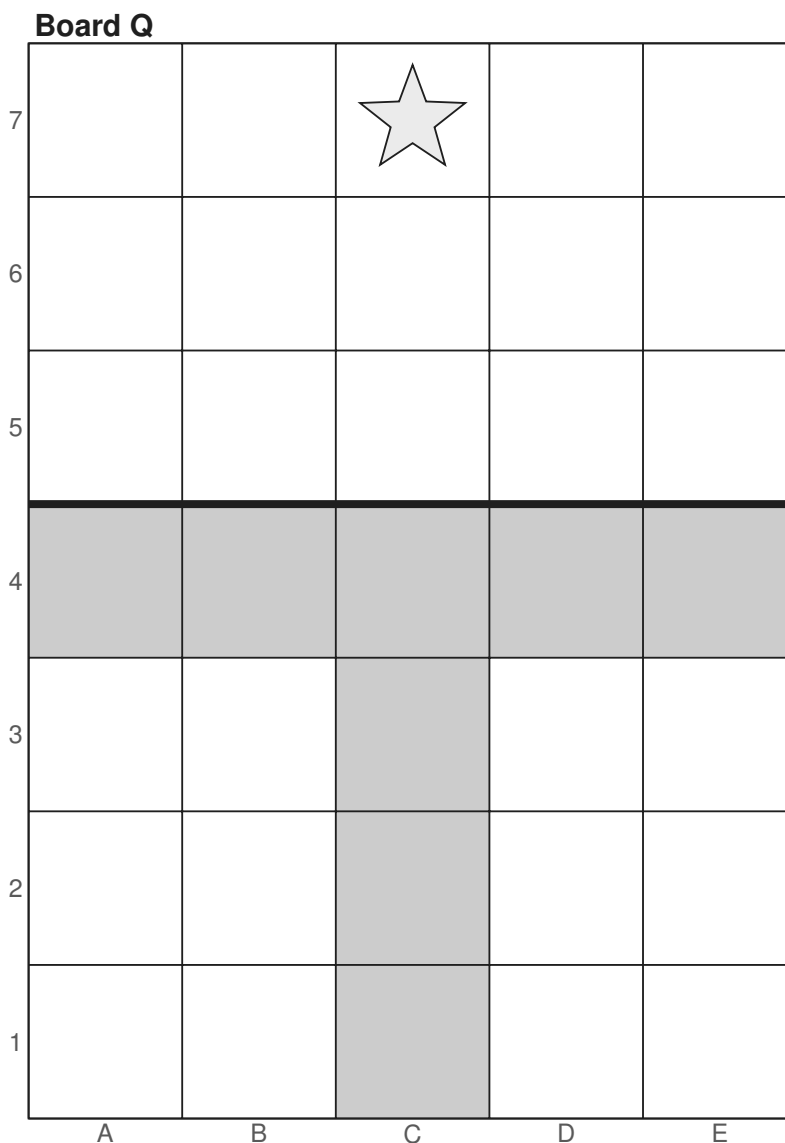
Board P

5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

Problem 10: On board P the star is worth 21, its neighbours 13, and the squares two steps away 8. Write the rule that gives every square of board P its value: _____. Then try to find a jump on board P that makes the total go up: try jumps toward the star, away from it, and sideways past it. Write what you find: _____

Problem 11: On board P, find the three home squares with the largest values and write their total: _____. Write the value of the star: _____. Put pennies on those three squares, play, and write the highest row any peg reached: _____. Now write the total of the four squares of shape I: _____, of shape L: _____ and of shape M: _____. Tell the adult what these numbers say about any three pegs, and what shape I shows.

Problem 12: Board Q has eight pegs: five along the river and three below the middle one. The whole table works on one board. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Agree on each jump before anyone makes it, and take turns writing the jumps down: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

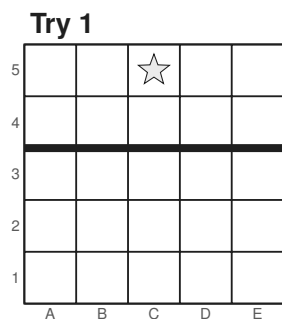
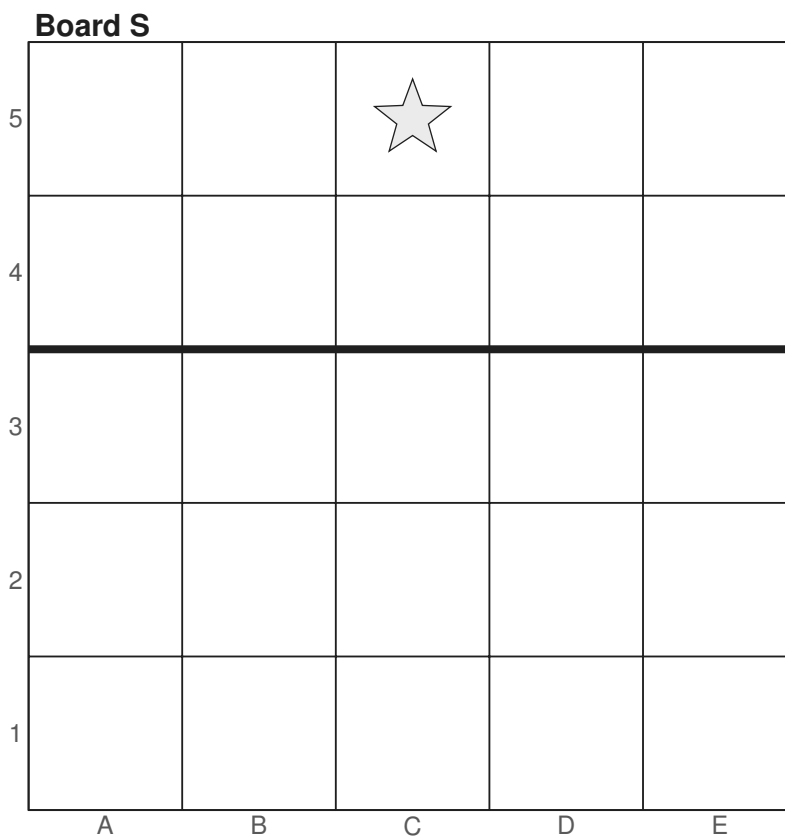
Jump 4: from _____ over _____ to _____

Problem 13: Board R shows the values that go with the star at C7. Write the total of the seven most valuable home squares: _____ and the value of the star: _____. Then write the total of the eight gray squares, the army of board Q: _____. Say what these three numbers tell you about armies of seven pegs and about the army of board Q, and write it in one or two sentences.

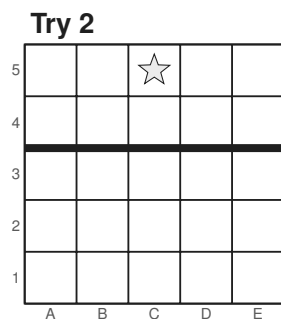
Board R

7	21	34	55 	34	21
6	13	21	34	21	13
5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

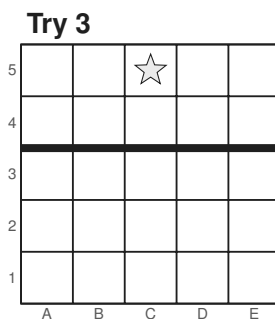
Problem 14 (Extra): Board S is empty. Put only three pennies on home squares of your choice and try to get a peg to the star at C5. Try at least four different placements. Record each one on a small board below: draw a dot in each square you used and write the highest row number any peg reached.



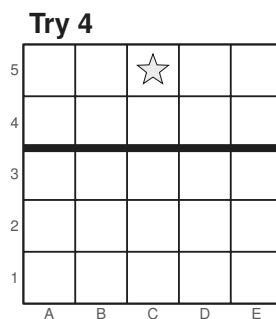
highest row: _____



highest row: _____

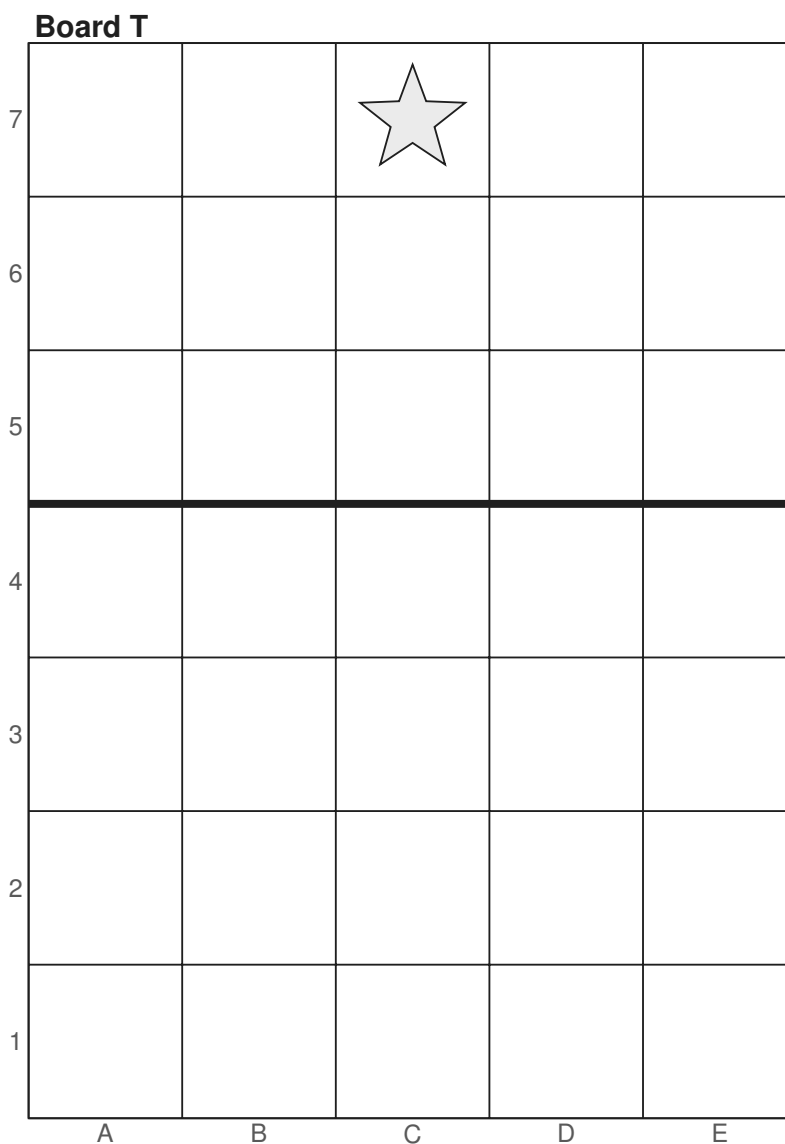


highest row: _____



highest row: _____

Problem 15 (Extra): Board T is empty. Find a set of eight home squares, different from the shape of board Q, from which a peg can reach the star at C7. Test it with pennies. Draw a dot in each of the eight squares of a set that works, and list its seven jumps.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

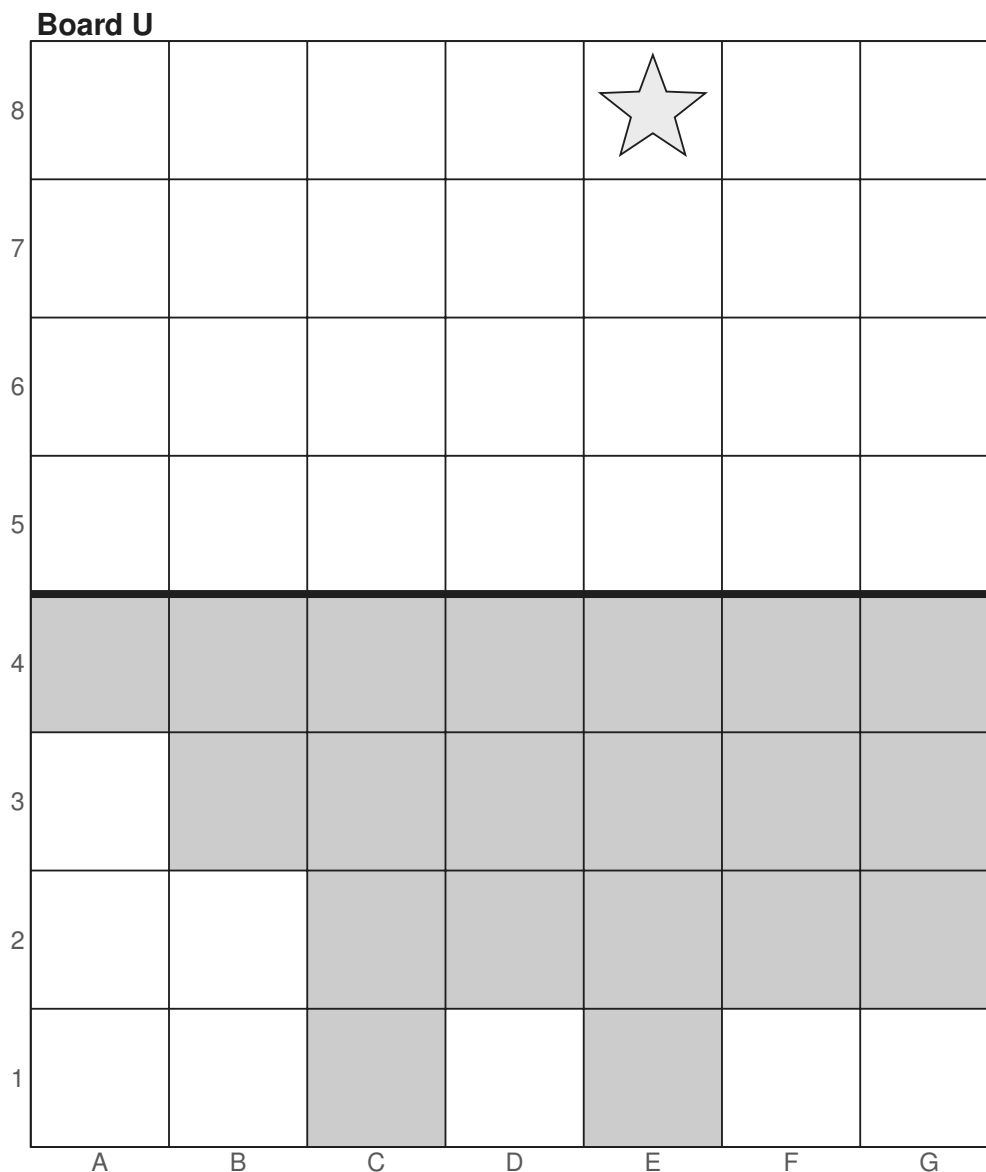
Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Problem 16 (Extra): Board U has twenty pegs, and its star is in the fourth row across the river, at E8. The whole table works on one board. Put pennies on the gray squares and find nineteen jumps that put a peg on the star. Before you start, decide which peg should be the last one to move and which two squares must hold pegs just before the final jump. Write the jumps down as you make them, on the lines after the board.



1: ____ over ____ to ____
 2: ____ over ____ to ____
 3: ____ over ____ to ____
 4: ____ over ____ to ____
 5: ____ over ____ to ____
 6: ____ over ____ to ____
 7: ____ over ____ to ____

8: ____ over ____ to ____
 9: ____ over ____ to ____
 10: ____ over ____ to ____
 11: ____ over ____ to ____
 12: ____ over ____ to ____
 13: ____ over ____ to ____
 14: ____ over ____ to ____

15: ____ over ____ to ____
 16: ____ over ____ to ____
 17: ____ over ____ to ____
 18: ____ over ____ to ____
 19: ____ over ____ to ____